

We have
crossed
2.5 million
hits per month !



Popular sections



Circuit Lab

Hundreds of circuit diagrams of different projects, for FREE



Top 100

Most popular technology websites, measured by their hits



Events

A comprehensive list of technical events, in India and abroad

electronicsforu.com

A comprehensive portal on electronics

Check it out today!

So consider both sides before deciding.

A phased approach is always a smart and cost-effective way to transit. The first phase of migration should be to establish the IPv6 transition team to define the team action plan. The following actions need to be taken then:

Network readiness assessment.

Security should always be the top priority. If IPv6 devices are already active in a network, a gaping hole in security might exist. Therefore assessing your network and identifying IPv6-enabled equipment should be the first step. This also includes risk and cost analysis.

Doing this will also allow to deduce how much new equipment are to be purchased. Most of your recent purchases have a high probability of being IPv6-ready. Most companies like ZTE, Cisco and Huawei have been adding IPv6 capabilities to their routers for some time now.

Moreover, since compatibility with IPv6 networking is mainly a software or firmware issue, a lot of hardware might not require replacement. Timely analysis in this regard might save a lot of funds.

Create a software inventory. Just like software or firmware updates for routers and other networking gear, there is also the question of the operating systems being used in an IT ecosystem supporting the new protocol. So one needs to make sure that all the software is inventoried. For instance, Microsoft Windows 7 can already handle IPv6 traffic, but Windows XP SP2 cannot. So computers running Windows XP SP2 needs to have the IPv6 part installed and activated manually via command line:

```
c:\> netsh interface ipv6 install
```

"The end justifies the means." A common mistake while setting the goal is to confuse between end goals and mean goals. While end goals define outcomes where one cannot compromise, mean goals form the paths to reach the end goal. First, one should understand the benefits that the company hopes to derive from moving to IPv6. Perhaps, it might have

important clients and business partners on the IPv6 network. Or the company hopes to grow its business on the new Internet.

Whatever it be, understanding this will help to decide how to transition. Early adopters may phase in IPv6 across the enterprise and begin application development based on IPv6. On the other hand, there might be companies that block all the IPv6 traffic from their networks and prolong deployment until they deem it to be robust enough. And then there are companies in between which while allowing IPv6 traffic through their firewall, will still shy away from investing heavily until all the issues are sorted out.

Develop a migration plan. Which-ever criteria a company belongs to, ultimately IPv6 will be embraced. It should ensure that the transition is done smoothly and steadily. One can begin by designing trial scenarios and then testing them. This requires developing migration scenarios and test cases, conducting the tests and evaluating the test results. If the test results are positive and successful transition is achieved in the test case, one can go ahead.

Initiate IPv6 transition. The initial transition should be restricted to one section only. This allows easier analysis and problem-solving. It also allows to control the risk involved in case of a mishap. Once the transition is complete, the service should be monitored continuously and all IPv6 operational gaps identified. Note down the new security measures needed to filter and monitor the IPv6 traffic. Security measures implemented for IPv4 should also be in place for IPv6 traffic.

After the networking hardware is upgraded or installed, software and firmware enabled and security tools for the new network configured, you are ready to go. Simply allow the IPv6 traffic to start flowing across the network and marvel at the ingenuity of the next-generation Internet! ●

The author is a tech correspondent at Efy